

Iron delocalization occurs during ischemia and persists on reoxygenation of skeletal muscle

JERRY J. C. CHIAO, RICHARD E. KIRSCHNER, and GARY A. FANTINI

NEW YORK, NEW YORK

Attenuation of oxidative reperfusion injury in skeletal muscle by the administration of iron-chelating compounds suggests that ischemia-reperfusion is associated with delocalization of iron with subsequent catalysis of hydroxyl radical generation. To test this hypothesis we examined the extent of iron liberation and lipid peroxidation in a well-established model of high-grade partial hindlimb ischemia and reperfusion. Laparotomy was performed on heparinized male Sprague-Dawley rats with isolation of the infrarenal aorta. Resting membrane potential (E_m) and conjugated diene content in hindlimb skeletal muscle were determined along with plasma iron concentration and percent saturation of transferrin in five groups of animals. Baseline animals ($n = 6$) underwent a 30-minute postoperative stabilization period before data collection; Sham ischemia animals ($n = 10$) underwent aortic exposure without clamping for 120 minutes; ischemia animals ($n = 8$) underwent aortic clamping for 120 minutes; sham reperfusion animals ($n = 8$) underwent aortic exposure without clamping for 150 minutes; reperfusion animals ($n = 8$) underwent aortic clamping for 120 minutes followed by 30 minutes of reperfusion. Iron delocalization occurred during ischemia, as indicated by a significant rise in percent saturation of transferrin over that of the corresponding sham group ($35\% \pm 2\%$ vs $25\% \pm 2\%$; $p < 0.05$) and persisted during reperfusion ($39\% \pm 5\%$ vs $27\% \pm 3\%$; $p < 0.05$). Depolarization of resting E_m was noted during ischemia (-75.7 ± 1.7 mV vs -92.6 ± 0.4 mV in the corresponding sham group; $p < 0.01$), with only partial repolarization demonstrated after reperfusion (-82.2 ± 1.7 mV; $p < 0.01$ vs all other groups). Reperfusion was also associated with a significant increase in skeletal muscle content of conjugated dienes ($p < 0.05$ vs all other groups). These data indicate that iron delocalization occurs during skeletal muscle ischemia and reperfusion and is associated with significant membrane dysfunction and postischemic lipid peroxidation. A role for iron in the catalysis of free radical-mediated oxidative reperfusion injury in skeletal muscle is strongly supported. (J Lab Clin Med 1994;124:432-8)

Abbreviations: ATP = Adenosine Triphosphate; ATPase = Adenosine Triphosphatase; BL = Baseline E_m = transmembrane potential; IS = ischemia; $O_2^{\cdot -}$ = superoxide anion; $OH^{\cdot}$ = hydroxyl radical; RP = reperfusion; SI = sham ischemia; SR = sham reperfusion; TIBC = total iron-binding capacity; UIBC = unsaturated iron-binding capacity

From the Department of Surgery and Division of Vascular Surgery, The New York Hospital and Cornell University Medical College.

Submitted for publication Dec. 3, 1993; revision submitted Feb. 17, 1994; accepted March 7, 1994.

Reprint requests: Gary A. Fantini, MD, The New York Hospital-Cornell Medical Center, Suite F-1920, 525 East 68th St., New York NY 10021.

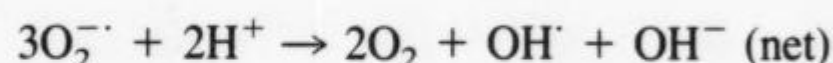
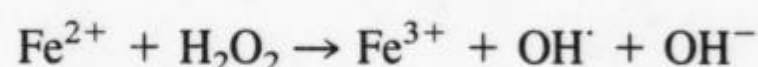
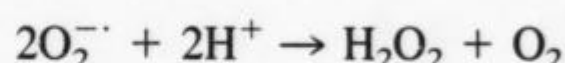
Copyright © 1994 by Mosby-Year Book, Inc.

0022-2143/94 \$3.00 + 0 5/1/56853

It has been recognized for over three decades that tissue hypoperfusion is associated with the appearance of increased levels of iron in the plasma.^{1,2} Recent observations have documented the liberation of iron into the circulation after reperfusion of ischemic myocardium³ and small intestine⁴ and into the urine after renal ischemia-reperfusion.⁵ Other studies have demonstrated delocalization of iron in the parenchyma of postischemic brain,^{6,7} myocardium,⁸

and kidney.⁹ Evidence documenting the participation of iron in the pathogenesis of tissue injury after ischemia-reperfusion is provided by studies in which deferoxamine and other iron-chelating agents attenuate postischemic injury in a variety of organ systems.^{5,10-13} Several reports have documented lack of efficacy in this regard with the iron-loaded compound ferrioxamine,^{11,13,14} suggesting that the primary protective mode of action of deferoxamine is related to its ability to bind iron. These results implicate delocalized iron in the pathogenesis of tissue injury accompanying ischemia and reperfusion.

A primary mechanism through which iron is thought to participate in injury to reoxygenated tissues is via catalysis of OH[•] formation by the Haber-Weiss reaction:



In this schema, iron functions as both a redox reagent, accepting electrons from O[•], and as a Fenton catalyst, promoting the formation of OH[•] from H₂O₂. Hydroxyl radical is an extremely reactive oxygen metabolite capable of initiating chain reactions of membrane phospholipid peroxidation and is generally believed to be the free radical species most responsible for tissue damage. In the absence of a transition metal catalyst, the Haber-Weiss reaction proceeds too slowly to be of physiologic significance, thus suggesting a major role of ferrous iron in promoting tissue injury through this pathway. In addition to catalyzing hydroxyl radical production, iron may also contribute to oxidative reperfusion injury via the reaction of its hypervalent states, such as ferryl and perferryl ions, or via its reaction with lipid hydroperoxides formed during lipid peroxidation, the latter leading to generation of alkoxy and peroxy radicals that are themselves capable of propagating lipid peroxidation reactions.¹⁵ Consequently there are several pathways through which delocalized iron may participate in the generation of tissue injury at reperfusion including some that may be independent of hydroxyl radical formation.

Smith et al.¹³ have demonstrated that the iron-binding compounds deferoxamine and apotransferrin attenuate the rise in microvascular permeability produced by ischemia and reperfusion of skeletal muscle in a rat model. In addition, Fantini and Yoshioka¹⁶ have documented that free deferoxamine inhibits lipid

peroxidation in skeletal muscle subjected to ischemia-reperfusion in a clinically relevant animal model. In the same model, similar protection is afforded by pentastarch-conjugated deferoxamine, indicating that extracellular iron-catalyzed oxidant production plays a critical role in reperfusion-induced skeletal muscle injury. Together these data indicate that the availability of catalytically active iron is a prerequisite to the genesis of postischemic skeletal muscle injury; however, delocalization of iron resulting from skeletal muscle ischemia-reperfusion has not yet been formally demonstrated. The present experiments were undertaken to examine the pathophysiology of oxidative skeletal muscle reperfusion injury, with specific attention devoted to the decompartmentalization and redistribution of iron.

METHODS

Animal preparation. Male Sprague-Dawley rats (250 to 340 gm) were obtained from a commercial vendor (Charles River Laboratories, Wilmington, Mass.) and allowed free access to food and water. Throughout the course of the experiments, animal care complied with the NIH Guide for the Care and Use of Laboratory Animals, and all animal protocols were approved by the Cornell University Medical College Institutional Animal Care and Use Committee. Animals were anesthetized by intraperitoneal injection of sodium pentobarbital (50 mg/kg) and placed on a warming table at 39° C. A tracheostomy tube was placed, and the animals were allowed to breathe room air. The right carotid artery and right external jugular vein were cannulated. Arterial blood pressure and heart rate were continuously monitored through the arterial line, and lactated Ringer's solution was infused at a constant rate of 1 ml/hr through the venous line. A thermister probe was inserted through the mouth and advanced into the esophagus for constant monitoring of core temperature. A period of 30 minutes was then allowed for stabilization of all animals. A midline laparotomy was subsequently performed, the aorta was isolated below the renal arteries, and heparin (1 U/gm body wt) administered intravenously. Bilateral hindlimb ischemia was induced by the application of an atraumatic microvascular clamp to the infrarenal aorta.

Experimental design. Rats were divided into five experimental groups: (1) BL animals (n = 6) were placed on a warming table for 30 minutes; (2) SI animals (n = 10) underwent aortic exposure without clamping for 120 minutes; (3) IS animals (n = 8) underwent infrarenal aortic clamping for 120 minutes; (4) SR animals (n = 8) underwent aortic exposure without clamping for 150 minutes; and (5) RP animals (n = 8) underwent aortic clamping for 120 minutes followed by 30 minutes of reperfusion. SI and SR groups served as controls to the IS and RP groups, respectively.

All data were gathered at one time point only, on completion of the experiment. Resting *E_m* was directly measured in gastrocnemius muscle fibers, after which the hindlimb adductor muscles were frozen in situ by using cold stainless

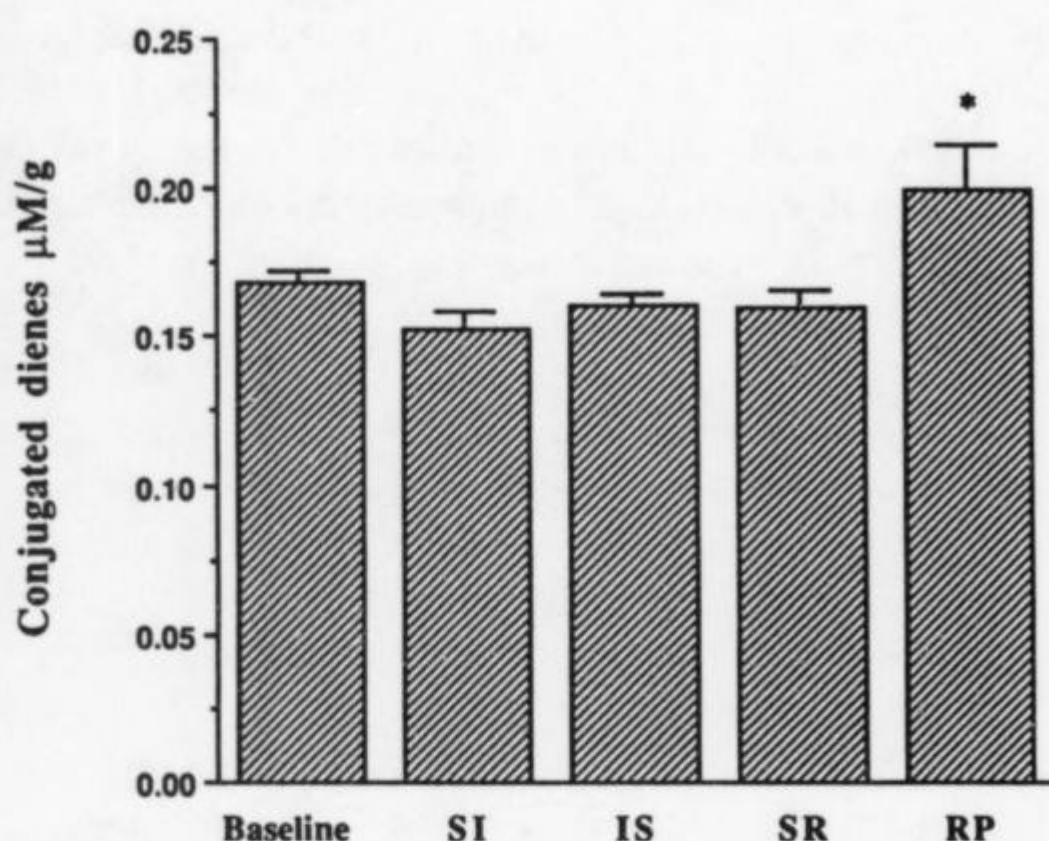


Fig. 1. Comparison of conjugated diene content of myocytes among BL, SI, IS, SR, and RP groups. Values are expressed as mean $\pm$ SEM. * $P < 0.05$ versus BL, SI, IS, and SR.

steel bars dipped in liquid nitrogen. These muscles were then excised, flash frozen in liquid nitrogen, and stored at -70°C for subsequent determination of conjugated diene content. Finally, arterial blood was drawn into a heparinized syringe for determination of plasma iron and UIBC. The TIBC was determined as the sum of plasma iron and UIBC; the percent saturation of transferrin was calculated as the ratio of plasma iron to TIBC.

Measurement of resting E_m . Resting E_m of gastrocnemius fibers was measured at the end of each experiment by using modified Ling-Gerard microelectrodes, as previously described.¹⁶

Analytic procedures. Conjugated diene content in skeletal muscle was measured by spectrophotometric detection. Total lipids were extracted from approximately 300 mg of muscle by the method of Bligh and Dyer.¹⁷ The lipids extracted in the chloroform layer were removed and evaporated to dryness under a stream of nitrogen gas at room temperature. These were then dissolved in heptane and the conjugated diene content determined by measuring the absorbance at 233 nm in a spectrophotometer (Beckman D-64; Beckman Instruments, Inc., Fullerton, Calif.) with a molar extinction coefficient of 28,000.¹⁸

Plasma iron level was measured by using the Rice modification¹⁹ of the method proposed by the International Committee for Standardization in Hematology.²⁰ The principle of the method is as follows: ferric iron in plasma is simultaneously dissociated from transferrin and reduced to the ferrous state, and the plasma proteins are precipitated with a protein precipitant containing hydrochloric, thioglycolic, and trichloroacetic acids. The ferrous iron in the clear extract of plasma is reacted with a solution containing sodium acetate and ferrozine, a chromogen. The resulting purple ferrous chromogen complex is quantitated spectrophotometrically at 562 nm and the iron concentration is calculated from the absorbance of an iron standard solution. The UIBC was measured by colorimetric determination with a Sigma

Table I. Comparison of skeletal muscle resting E_m

Group	$E_m(\text{mV})$
BL	-93.3 ± 0.6
SI	-92.6 ± 0.4
IS	$-75.7 \pm 1.7^*$
SR	-91.8 ± 0.4
RP	$-82.2 \pm 1.7^\dagger$

Values are expressed as mean $\pm$ SEM.

* $p < 0.01$ versus BL, SI, and RP.

† $p < 0.01$ versus BL, IS, and SR.

Diagnostics Kit (procedure no. 565; Sigma Chemical Co., St. Louis, Mo.). At alkaline pH, ferrous ions are added to plasma to bind specifically with transferrin at unsaturated iron-binding sites. Remaining unbound ferrous ions are measured with the ferrozine reaction. The difference between the amount of unbound iron and the total amount added to the plasma is equivalent to the quantity bound to transferrin. This is the UIBC. The TIBC of plasma equals plasma iron plus UIBC.

Statistical analysis. Data are presented as mean $\pm$ SEM. The difference between two group means was determined by one-way analysis of variance and, when appropriate, by Duncan's multiple range test. A p value of 0.05 or less was considered to be significant in all instances.

RESULTS

Infrarenal aortic clamping for 120 minutes resulted in significant depolarization of the resting E_m in the IS group ($p < 0.01$) as compared with BL and SI (Table I). After 30 minutes of reperfusion, partial membrane repolarization was noted ($p < 0.01$ vs IS); however, the membrane was still significantly depolarized as compared with BL and SR ($p < 0.01$). No significant difference in resting E_m was noted between the BL group and the SI and SR groups.

Significant lipid peroxidation was evident after 30 minutes of reperfusion, as indicated by an increase in skeletal muscle conjugated diene content ($p < 0.05$ vs all other groups; Fig. 1). Tissue conjugated diene content remained unchanged during ischemia.

Significant increases in the percent saturation of transferrin and plasma iron were noted after 120 minutes of ischemia ($p < 0.05$ vs respective sham group, SI; Figs. 2 and 3). After 30 minutes of reperfusion, percent saturation of transferrin and plasma iron remained significantly elevated ($p < 0.05$ vs BL and SR groups). There were no differences in the TIBC between any of the groups.

The plasma iron levels in both sham groups (SI = $99.8 \pm 6 \mu\text{g}/100 \text{ ml}$ and SR = $100.7 \pm 8 \mu\text{g}/100 \text{ ml}$) were approximately 18% lower than the BL group ($122.6 \pm 12 \mu\text{g}/100 \text{ ml}$), although these differences were not statistically significant. This apparent hypo-

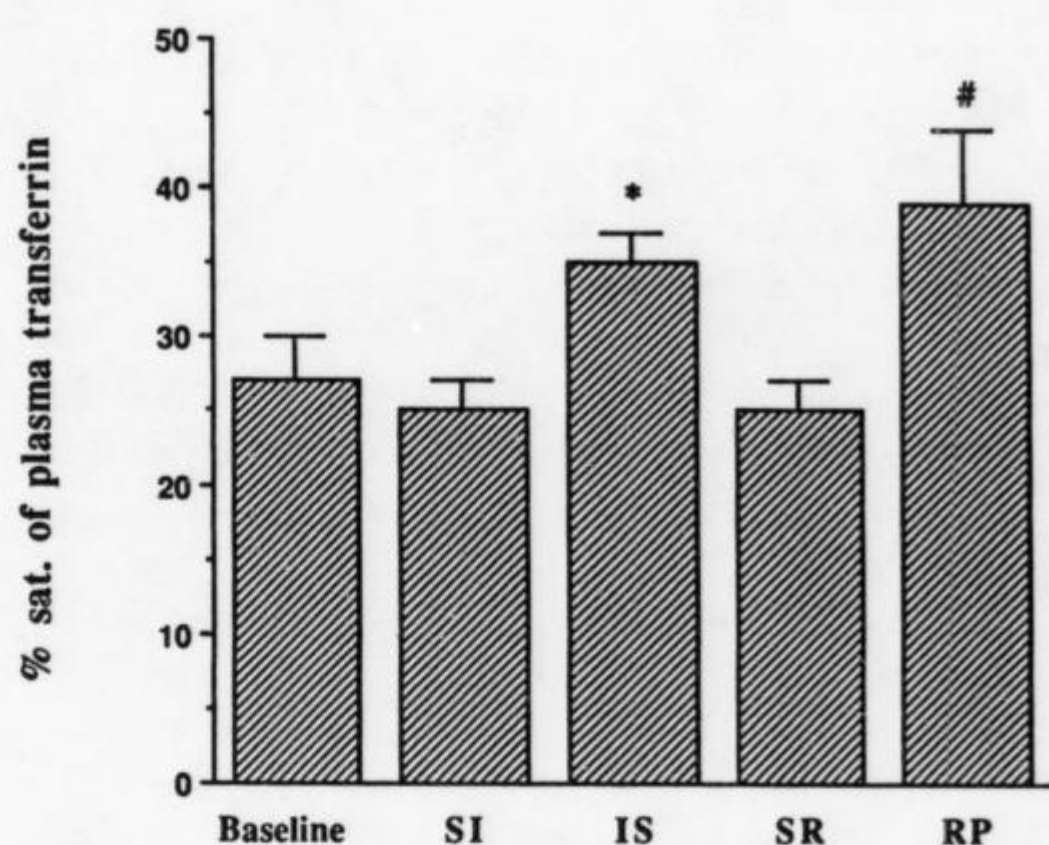


Fig. 2. Comparison of percent saturation of plasma transferrin among BL, SI, IS, SR, and RP groups. Values are expressed as mean $\pm$ SEM. * $p < 0.05$ versus SI; # $p < 0.05$ versus BL and SR.

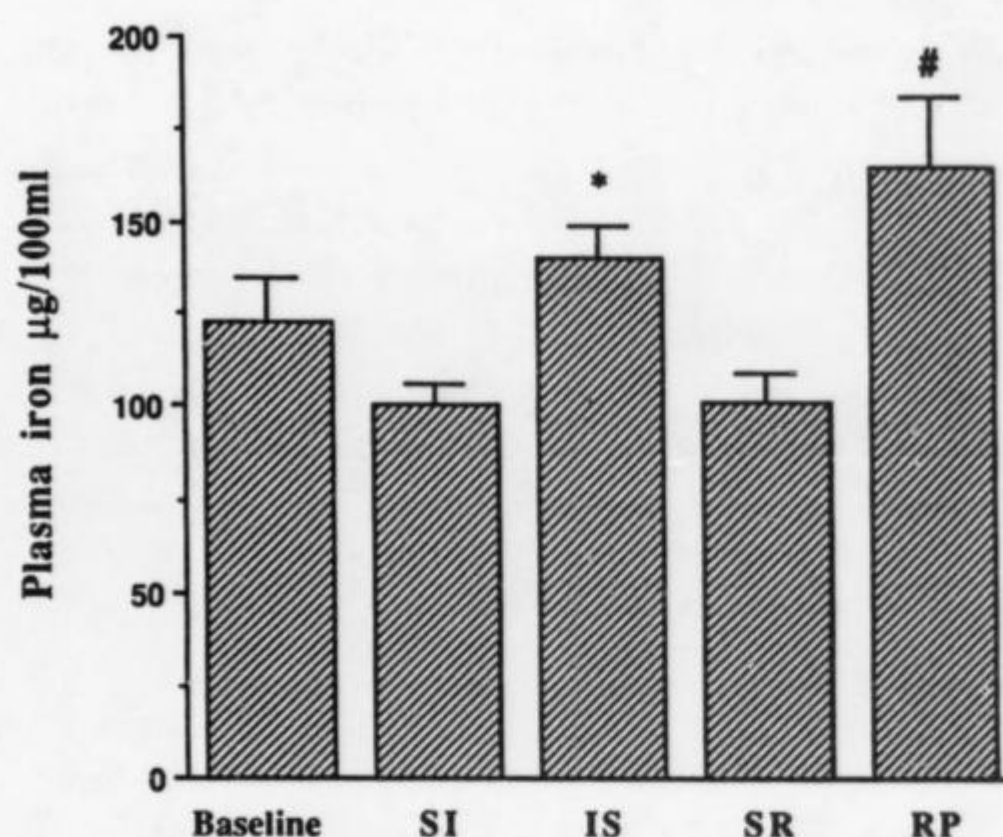


Fig. 3. Comparison of plasma iron among BL, SI, IS, SR, and RP groups. Values are expressed as mean $\pm$ SEM. * $p < 0.05$ versus SI; # $p < 0.05$ versus BL and SR.

ferremic response occurring only in the sham groups parallels an observation by Robinson and Hedlund in which rats subjected to laparotomy only demonstrated a similar decrease in plasma iron levels.⁴ In addition, Mazur et al.¹ reported that the eventual increase in plasma iron noted in dogs subjected to severe hypotension was preceded by a transient decrease in plasma iron early in the course of hemorrhagic shock. Janoff et al.² reported a similar response after relief of experimental mesenteric artery occlusion in dogs. Collectively these observations suggest that hypoferremia represents a generalized and early response to trauma and stress and serves to amplify the observed increases in plasma iron accompanying ischemia and reperfusion.

DISCUSSION

Numerous studies have implicated reactive oxygen metabolites in the pathogenesis of injury produced by the reperfusion of ischemic skeletal muscles.^{16,21,22} Although considerable attention has been devoted to the role of O_2^- in this regard, it is unlikely that O_2^- directly mediates ischemia-reperfusion-induced cellular injury because of its relatively low reactivity under physiologic conditions and its consequent inability to peroxidize membrane lipids.¹⁵ Moreover, the efficiency of endogenous superoxide dismutase likely ensures that the majority of O_2^- produced in reoxygenated tissues is rapidly consumed in the formation of hydrogen peroxide.²³ As a result, it is now widely believed that oxidative injury in postischemic tissues is a result of the action of the more highly reactive hydroxyl radical, which may be generated from O_2^- and hydrogen peroxide via the iron-catalyzed Haber-

Weiss reaction. A role for $\text{OH}^\cdot$ in the genesis of postischemic skeletal muscle injury is likely, since administration of the hydroxyl radical scavenger dimethyl sulfoxide protects against ischemia-reperfusion-induced injury in canine gracilis muscle.²² Recent studies have demonstrated that the iron chelator deferoxamine significantly attenuates lipid peroxidation and the rise in microvascular permeability produced by ischemia and reperfusion of the rat hindlimb,^{13,16} supporting the hypothesis that iron-catalyzed oxidant production plays a critical role in the pathogenesis of skeletal muscle reperfusion injury.

In vitro studies clearly demonstrate that ferrous iron is capable of promoting lipid peroxidation.²⁴ Under normal circumstances, the concentration of unbound iron available to catalyze the formation of $\text{OH}^\cdot$ in tissues is extremely low; however, after an ischemic insult substantial quantities of stored iron may be liberated, thus providing a suitable catalyst for the Haber-Weiss reaction. It is known that iron released into the circulation from intracellular binding sites is immediately bound to the transport protein transferrin, and thus virtually no "free" iron exists in plasma.²⁴ In the 1950s Mazur and associates^{1,25,26} demonstrated that iron was released during the global hypoperfusion of hemorrhagic shock in quantities sufficient to saturate the transferrin binding capacity. A number of more recent studies have indicated that ischemia-reperfusion of isolated organs may also result in iron delocalization. Krause et al.⁷ demonstrated an increase in the concentration of low molecular weight iron species in the canine brain after cardiac arrest and resuscitation. Similarly, Holt et al.⁸ noted increased levels of low molecular weight iron species in homogenates of isch-

emic myocardium. The extracellular release of iron has been reported by Paller and Hedlund,⁵ who demonstrated a 10-fold increase in urinary free iron after renal ischemia-reperfusion, and by Robinson and Hedlund,⁴ who noted a substantial rise in transferrin-bound iron after reperfusion of the ischemic small intestine. Our results demonstrate a significant rise in plasma iron and in percent saturation of transferrin occurring after 120 minutes of high-grade partial hindlimb ischemia and persisting after 30 minutes of reperfusion. These findings indicate the release of significant quantities of iron from storage sites into the circulation during hindlimb ischemia-reperfusion.

Although the source of delocalized iron was not specifically addressed by the present experiments, several studies suggest that a likely source is the storage protein ferritin.^{25,27} Release of iron from ferritin requires the reduction of insoluble ferric (Fe^{3+}) iron to the ferrous (Fe^{2+}) state. The development of an acidic reducing environment during ischemia would favor the reduction and release of ferritin-bound iron, although the exact nature of the physiologic reductants involved is uncertain. Recent evidence suggests that O_2^- generated by the activity of xanthine oxidase can mediate the release of iron from ferritin.²⁷⁻²⁹ Moreover, in the presence of xanthine and xanthine oxidase, ferritin has been noted to promote peroxidation of phospholipid liposomes.²⁷ These findings may account in part for the ability of superoxide dismutase to protect against oxidative injury in postischemic tissues despite the low oxidative potential of O_2^- itself. Nevertheless, the role of ferritin as a source of the iron released during ischemia-reperfusion remains speculative, and there is conflicting evidence of its ability to serve as a Haber-Weiss catalyst under physiologic conditions.³⁰ Other potential sources of catalytically active iron include iron-sulfur (Fe-S) clusters associated with mitochondrial aconitase^{31,32} and the structurally similar iron-responsive element binding protein,³³ both of which are believed to be sensitive to oxidative attack by O_2^- .

Iron delocalization during ischemia and reperfusion in our experiments was associated with skeletal muscle membrane dysfunction, as indicated by depolarization of the cell membrane after 120 minutes of aortic occlusion with only partial repolarization after 30 minutes of reperfusion. Previous experiments from this laboratory have indicated that the cellular membrane injury accompanying ischemia and reperfusion may occur through fundamentally different mechanisms, as only the latter is attenuated by free radical scavengers³⁴ and iron chelators.¹⁶ These findings suggest that the membrane dysfunction observed during reperfusion may involve iron-catalyzed oxidant pro-

duction, whereas that noted during ischemia is independent of such pathways. Support for this hypothesis is provided by our data indicating that lipid peroxidation, as indicated by an increase in skeletal muscle conjugated dienes, occurred only in the reperfusion group. This finding is consistent with reports from another laboratory in which lipid peroxidation detected by measurement of conjugated dienes was evident only after reperfusion of ischemic canine skeletal muscle.^{35,36}

As the cell membrane defines the boundary between the intracellular and extracellular compartments, it is uniquely susceptible to attack by reactive oxygen metabolites generated during reperfusion in either compartment. Direct measurement of the resting E_m provides a functional assessment of the status of the cell membrane, while determination of muscle-conjugated dienes provides a complementary index of structural oxidative membrane injury. A detailed discussion of the generation and maintenance of transmembrane ionic gradients and physiologic factors influencing resting E_m can be found elsewhere.^{37,38} The cell membrane is normally relatively impermeable to Na^+ , and the existence of an intracellular negative charge is generated by the continuous efflux of K^+ down a concentration gradient, from an area of relatively high concentration within the cell to an area of much lower concentration outside the cell. Maintenance of this K^+ diffusion gradient, and hence of the resting E_m , is due primarily to the action of the membrane-associated, ATP-dependent protein complex (Na^+/K^+ ATPase), which pumps Na^+ out of the cell and K^+ into the cell in a ratio of 3:2. The electrogenic nature of this pump also contributes to the resting E_m , but only to a very limited extent. Thus, possible mechanisms through which factors influencing resting E_m might act include alteration of the relative membrane permeabilities of Na^+ and K^+ ($p\text{Na}/p\text{K}$) or of the transmembrane K^+ concentration gradient by altering the rate of Na^+/K^+ pumping.

Maridonneau et al.³⁹ have demonstrated that superoxide anion at low concentrations can increase the passive K^+ permeability of the cell membrane, independent of Na^+/K^+ ATPase activity. Thus it is possible that small amounts of O_2^- generated during high-grade partial ischemia in our experiments induced similar changes in passive membrane permeability, with stagnant blood flow allowing for an accumulation of K^+ in the extracellular space and subsequent depolarization of the resting E_m . Recent experiments have shown that the activity of the Na^+/K^+ ATPase is also remarkably sensitive to oxidative stress.^{40,41} Oxygen-derived free radicals generated as a consequence of reperfusion can markedly depress

Na⁺-K⁺ ATPase pump activity, resulting in a net K⁺ efflux and subsequent membrane depolarization. The mechanism of pump inhibition is likely related either to oxidant-induced changes in the lipid environment of the ATPase molecule or to a direct oxidative attack on the pump proteins themselves.⁴² Shattock and Matsuura,⁴¹ noting an absence of changes in passive membrane currents during ischemia-reperfusion, have postulated that inhibition of the Na⁺-K⁺ ATPase is the result of oxidative modification of the pump molecule. The increase in skeletal muscle conjugated dienes noted during reperfusion in our experiments, however, suggests that persistent membrane depolarization may have been the result of peroxidation of the membrane environment of the pump. This inference is supported by the work of Kramer et al.,⁴³ who demonstrated pump inhibition resulting from free radical-induced membrane lipid peroxidation in cardiac myocytes.

In summary, our data provide direct evidence that iron is liberated in substantial quantities during skeletal muscle ischemia and reperfusion in vivo and that such iron delocalization is accompanied by cell membrane dysfunction and postischemic phospholipid peroxidation. These findings are consistent with the hypothesis that delocalization of iron during ischemia primes the extracellular environment such that the ischemic tissues are susceptible to iron-mediated oxidative injury on reoxygenation.

We thank Ram Inamdar for technical support.

REFERENCES

1. Mazur A, Baez S, Shorr E. The mechanism of iron release from ferritin as related to its biological properties. *J Biol Chem* 1955;213:147-160.
2. Janoff A, Zweifach BW, Shapiro LR. Levels of plasma-bound iron in experimental shock in the rabbit and dog. *Am J Physiol* 1960;198:1161-5.
3. Chevion MYJ, Har-El R, Berenshtein E, Uretzky G, Kitrossky N. Copper and iron are mobilized following myocardial ischemia: possible predictive criteria for tissue injury. *Proc Natl Acad Sci USA* 1993;90:1102-6.
4. Robinson E, Hedlund B. Role of iron in ischemia and reperfusion [Abstract]. *Circ Shock* 1989;27:367.
5. Paller MS, Hedlund BE. Role of iron in postischemic renal injury in the rat. *Kidney Int* 1988;34:474-80.
6. Komara JS, Nayini NR, Bialick HA, et al. Brain iron delocalization and lipid peroxidation following cardiac arrest. *Ann Emerg Med* 1986;15:384-9.
7. Krause GS, Joyce KM, Nayini NR, et al. Cardiac arrest and resuscitation: brain iron delocalization during reperfusion. *Ann Emerg Med* 1985;14:1037-43.
8. Holt S, Gunderson M, Joyce K et al. Myocardial tissue iron delocalization and evidence for lipid peroxidation after two hours of ischemia. *Ann Emerg Med* 1986;15:1155-9.
9. Gower JD, Healing G, Fuller BJ, Green CJ. Iron redistribution and lipid peroxidation in the cold ischaemic kidney. *Adv Exp Med Biol* 1990;264:393-6.
10. Hernandez LA, Grisham MB, Granger DM. A role for iron in oxidant-mediated ischemic injury to intestinal microvasculature. *Am J Physiol* 1987;253(Gastrointest Liver Physiol):G49-G53.
11. Bolli R, Patel BS, Jeroudi MO, et al. Iron-mediated radical reactions upon reperfusion contribute myocardial "stunning." *Am J Physiol (Heart Circ Physiol)* 1990;259:H1901-11.
12. Braugher JM, Burton PS, Chase RL, et al. Novel membrane localized iron chelators as inhibitors of iron-dependent lipid peroxidation. *Biochem Pharmacol* 1988;37:3853-60.
13. Smith JK, Carden DL, Grisham MB, Granger DN, Korthuis RJ. Role of iron in postischemic microvascular injury. *Am J Physiol* 1989;256(Heart Circ Physiol 25):H1472-7.
14. Maruyama M, Pieper GM, Kalyanaraman B, Hallaway PE, Hedlund BE, Gross GJ. Effects of hydroxyethyl starch conjugated deferoxamine on myocardial functional recovery following coronary occlusion and reperfusion in dogs. *J Cardiovasc Pharmacol* 1991;17:166-75.
15. Halliwell B, Gutteridge JMC. The importance of free radicals and catalytic metal ions in human diseases. *Molec Aspects Med* 1985;8:89-193.
16. Fantini GA, Yoshioka T. Deferoxamine prevents lipid peroxidation and attenuates reoxygenation injury in postischemic skeletal muscle. *Am J Physiol* 1993;264(Heart Circ Physiol 33):H1953-9.
17. Bligh EG, Dyer WJ. A rapid method of total lipid extraction and purification. *Can J Biochem Physiol* 1959;37:911-7.
18. Watson BD, Busto R, Goldberg WJ, Santiso M, Yoshida S, Ginsberg MD. Lipid peroxidation in vivo induced by reversible global ischemia in rat brain. *J Neurochem* 1984;42:268-74.
19. Rice EW, Fenner HE. Study of the ICSH proposed reference method for serum iron assay: obtaining optically clear filtrates and substitution of ferrozine. *Clin Chim Acta* 1974;53:391-3.
20. Lewis SM. International committee for standardization in hematology: proposed recommendations for measurement of serum iron in human blood. *Am J Clin Pathol* 1971;56:543-5.
21. Korthuis RJ, Smith JK, Carden DL. Hypoxic reperfusion attenuates postischemic microvascular injury. *Am J Physiol* 1989;256(Heart Circ Physiol 25):H315-9.
22. Korthuis RJ, Granger DN, Townsley MI, Taylor AE. The role of oxygen-derived free radicals in ischemia-induced increases in canine skeletal muscle vascular permeability. *Circ Res* 1985;57:599-609.
23. Weiss SJ. Tissue destruction by neutrophils. *N Engl J Med* 1989;320:365-76.
24. Halliwell B, Gutteridge JMC. Oxygen toxicity, oxygen radicals, transition metals and disease. *Biochem J* 1984;219:1-14.
25. Mazur A, Green S, Saha A, Carleton A. Mechanism of release of ferritin iron in vivo by xanthine oxidase. *J Clin Invest* 1958;37:1809-17.
26. Green S, Mazur A. Relation of uric acid metabolism to release of iron from hepatic ferritin. *J Biol Chem* 1957;219:653-8.
27. Thomas CE, Morehouse LA, Aust SD. Ferritin and superoxide-dependent lipid peroxidation. *J Biol Chem* 1985;260:3275-80.
28. Biemond P, Swaak AJG, Beindorff CM, Koster JF. Superoxide-dependent and -independent mechanisms of iron mobilization from ferritin by xanthine oxidase. *Biochem J* 1986;239:169-73.
29. Bolann BJ, Ulvik RJ. Release of iron from ferritin by xanthine oxidase: role of the superoxide radical. *Biochem J* 1987;243:55-9.
30. Kviety PR, Inauen W, Bacon BR, Grisham MB. Xanthine oxidase-induced injury to endothelium: role of intracellular iron and hydroxyl radical. *Am J Physiol* 1989;257(Heart Circ Physiol 26):H1640-6.

31. Beinert H. Recent developments in the field of iron-sulfur proteins. *FASEB J* 1990;4:2483-91.
32. Gardner PR, Fridovich I. Superoxide sensitivity of the *Escherichia coli* aconitase. *J Biol Chem* 1991;266:19328-33.
33. Haile DJ, Rouault TA, Tang CK, Chin J, Harford JB, Klausner RD. Reciprocal control of RNA-binding and aconitase activity in the regulation of the iron-responsive element binding protein: role of the iron-sulfur cluster. *Proc Natl Acad Sci USA* 1992;89:7536-40.
34. Yokota J, Minei JP, Fantini GA, Shires GT. Role of leukocytes in reperfusion injury of skeletal muscle after partial ischemia. *Am J Physiol* 1989;257(Heart Circ Physiol 26):H1068-75.
35. Harris K, Walker PM, Mickle AG, et al. Metabolic response of skeletal muscle to ischemia. *Am J Physiol* 1986;250(Heart Circ Physiol 19):H213-20.
36. Lindsay T, Walker PM, Mickle DAG, Romaschin AD. Measurement of hydroxyconjugated dienes after ischemia-reperfusion in canine skeletal muscle. *Am J Physiol* 1988;254:H578-83.
37. Fantini GA, Roberts JP, Lowry SF, et al. Effect of hormonal and substrate backgrounds on cell membrane function in normal males. *J Appl Physiol* 1987;63:1107-13.
38. Fantini GA, Zadeh BJ, Chiao J, Krieger KH, Isom OW, Shires GT. Effect of hypothermia on cellular membrane function during low-flow extracorporeal circulation. *Surgery* 1987;102:132-9.
39. Maridonneau I, Braquet P, Garay RP. Na⁺ and K⁺ transport damage induced by oxygen free radicals in human red cell membranes. *J Biol Chem* 1983;258:3107-13.
40. Kako K, Kato M, Matsuoka T, Mustapha A. Depression of membrane-bound Na⁺-K⁺-ATPase activity induced by free radicals and by ischemia of kidney. *Am J Physiol* 1988;254(Cell Physiol 23):C330-7.
41. Shattock MJ, Matsuura H. Measurement of Na⁺-K⁺ pump current in isolated rabbit ventricular myocytes using the whole-cell voltage-clamp technique: inhibition of the pump by oxidant stress. *Circ Res* 1993;72:91-101.
42. Xie Z, Wang Y, Askari A, Huang W-H, Klaunig JE. Studies on the specificity of the effects of oxygen metabolites on cardiac sodium pump. *J Mol Cell Cardiol* 1990;22:911-20.
43. Kramer JH, Tong Mak I, Weglicki WB. Differential sensitivity of canine sarcolemmal and microsomal enzymes to inhibition by free radical-induced lipid peroxidation. *Circ Res* 1984;55:120-4.